

Solution Blueprint: Aerospace Service Bulletin Warranty Adjudication Agent (Simplified Sample)

Automated Warranty Claim Adjudication for Commercial Aircraft Fleets

Multi-Agent Cognitive Architecture with Human-in-the-Loop A2UI Surfaces on Google Cloud

SIMPLIFIED ARCHITECTURAL SAMPLE & REFERENCE PATTERN

This blueprint is a simplified, synthetic architectural sample created to illustrate a multi-agent cognitive pattern with interactive human-in-the-loop A2UI surfaces on Google Cloud. All commercial contract terms, airline identifiers, service bulletin codes, and labor reimbursement policies are generic, synthetic examples created for reference purposes and do not represent any proprietary customer IP or production data.

Executive Document Information

Attribute	Specification
Client / Domain:	Commercial Aircraft Manufacturing & Fleet Support
Solution Category:	Warranty Claim Adjudication & Financial Reconciliation Automation
Technology Stack:	Google Agent Development Kit (ADK) 2.0, Gemini 3.5 Flash, Vertex AI Agent Engine, A2UI Protocol v0.9, Gemini Enterprise
Data Architecture:	Cloud SQL PostgreSQL Ledger + Cloud Storage MD5 Cache + Master Claims Database
Hosting Region:	Google Cloud europa-west1 (Belgium) / Runtime: Vertex AI Reasoning Engine
Document Classification:	Enterprise Solution Blueprint & Validated Reference Architecture (Sample)
Status:	Generic Proof-of-Concept & Reference Architecture Pattern
Version:	1.0.0 (Sanitized & Simplified Sample Distribution)

Executive Summary & Solution Highlights

Google Cloud Architecture Statement:

The **Aerospace Service Bulletin (SB) Warranty Adjudication Agent** is a reference architecture engineered to automate and govern the warranty reimbursement lifecycle for commercial aircraft fleets. When manufacturers publish Service Bulletins for technical modifications, airline operators perform maintenance retrofits and submit warranty reimbursement claims for documented labor and materials.

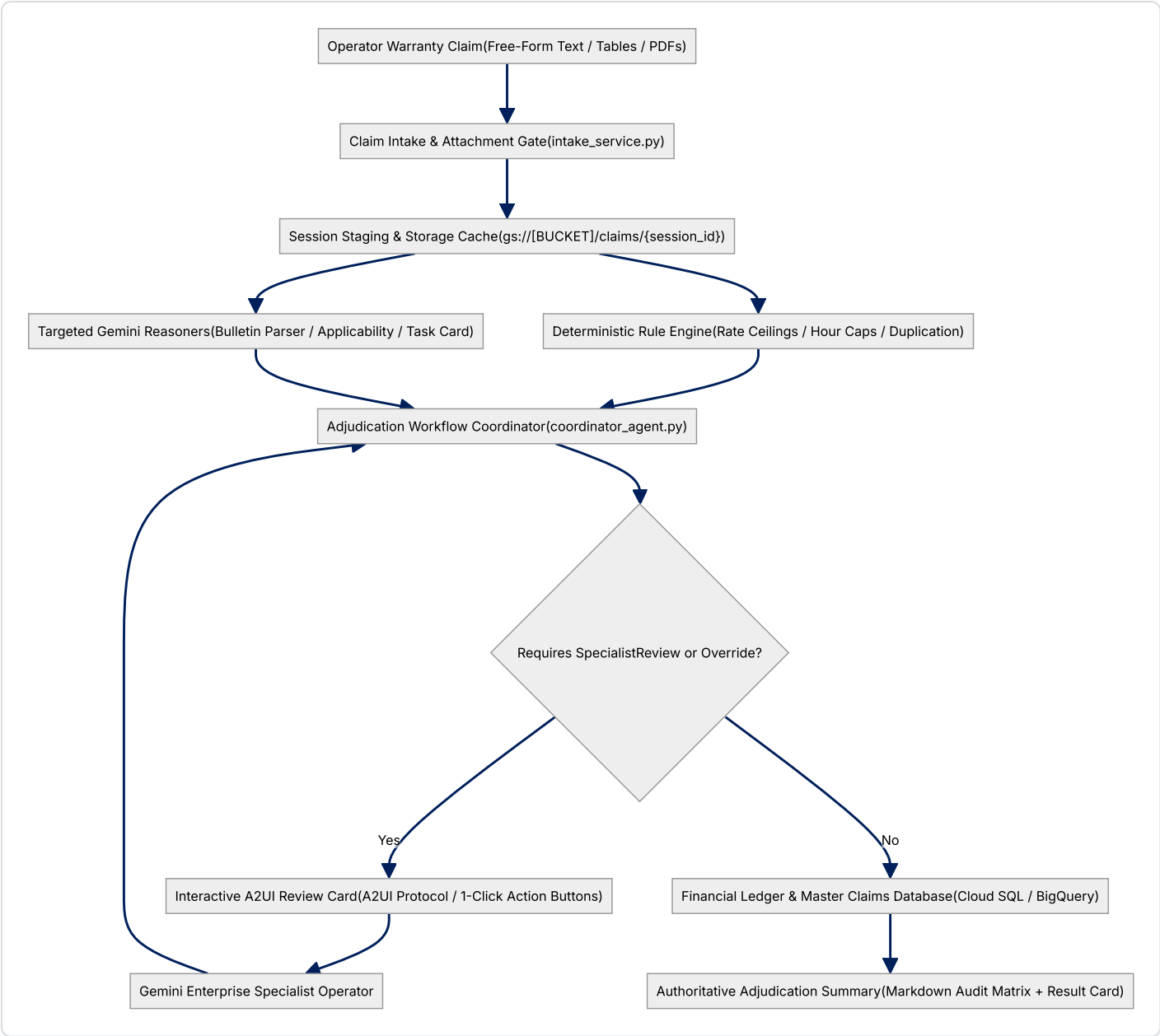
Adjudicating warranty claims requires multi-factor compliance verification:

- Airframe Applicability:** Confirming claimant aircraft Manufacturer Serial Numbers (MSNs) fall within the designated applicability ranges.
- Warranty & Campaign Policy:** Validating airframe warranty status or eligibility under a Commercial Campaign Notice (CCN).
- Task Card Verification:** Cross-referencing maintenance records to confirm aircraft registration, embodiment dates, and recorded technician hours.
- Labor Rate & Hours Reconciliation:** Applying contractual labor rate ceilings and enforcing procedure hour allowances.
- Anti-Duplication Ledger:** Preventing duplicate claim submissions against historical transaction records.

6. **Human-in-the-Loop Governance:** Providing warranty specialists with interactive A2UI review cards directly in Gemini Enterprise.

 15-Second Adjudication	 Deterministic Hours Reconciliation	 Anti-Duplication Protection	 Interactive A2UI Surfaces
From 60 min to <15s per claim dossier	Dual-Policy Rule Engine Deterministic hours cap	Historical Ledger Audit Zero duplicate payouts	Gemini Enterprise & A2UI 1-Click Action Buttons

1. Multi-Agent Cognitive Architecture



2. Streamlined Subagent Specifications

The solution organizes capabilities into five specialized agents and deterministic components:

Agent / Component	Execution Engine	Primary Responsibility	Validation / Decision Rule
warranty_coordinator	ADK Workflow	Session lifecycle, agent routing, and final response packaging.	Maintains audit context, manages execution flow, and streams UI events.
claim_intake_agent	Gemini 3.5 Flash	Parses unstructured emails, tables, and PDFs into structured claim models.	Extracts claimant ID, tail number, serial number, bulletin reference, and claimed amounts.
bulletin_analyzer_agent	Gemini 3.5 Flash	Evaluates service bulletin requirements and airframe applicability.	Verifies serial number effectivity ranges (MSN 10001 thru 10500) and standard hour allowances.
reconciliation_engine	Deterministic Engine	Reconciles documented actual hours against claimed and allowed values.	Applies contractual labor rate caps and calculates reimbursable financial totals.
a2ui_review_surface	A2UI Protocol v0.9	Renders interactive approval cards with 1-click action buttons.	Dispatches structured operator decisions (APPROVE , REJECT , HOLD , PROCEED).

3. Mathematical Labor Reconciliation Engine

The agent enforces deterministic mathematical formulations to compute reimbursable hours and credits:

$$\text{Reconciled Hours} = \min(\text{Documented Actual Hours}, \text{Claimed Hours}, \text{Allowed Bulletin Hours})$$

$$\text{Applied Labor Rate} = \min(\text{Claimed Rate}, \text{Contractual Negotiated Rate})$$

$$\text{Total Financial Credit} = \text{Reconciled Hours} \times \text{Applied Labor Rate}$$

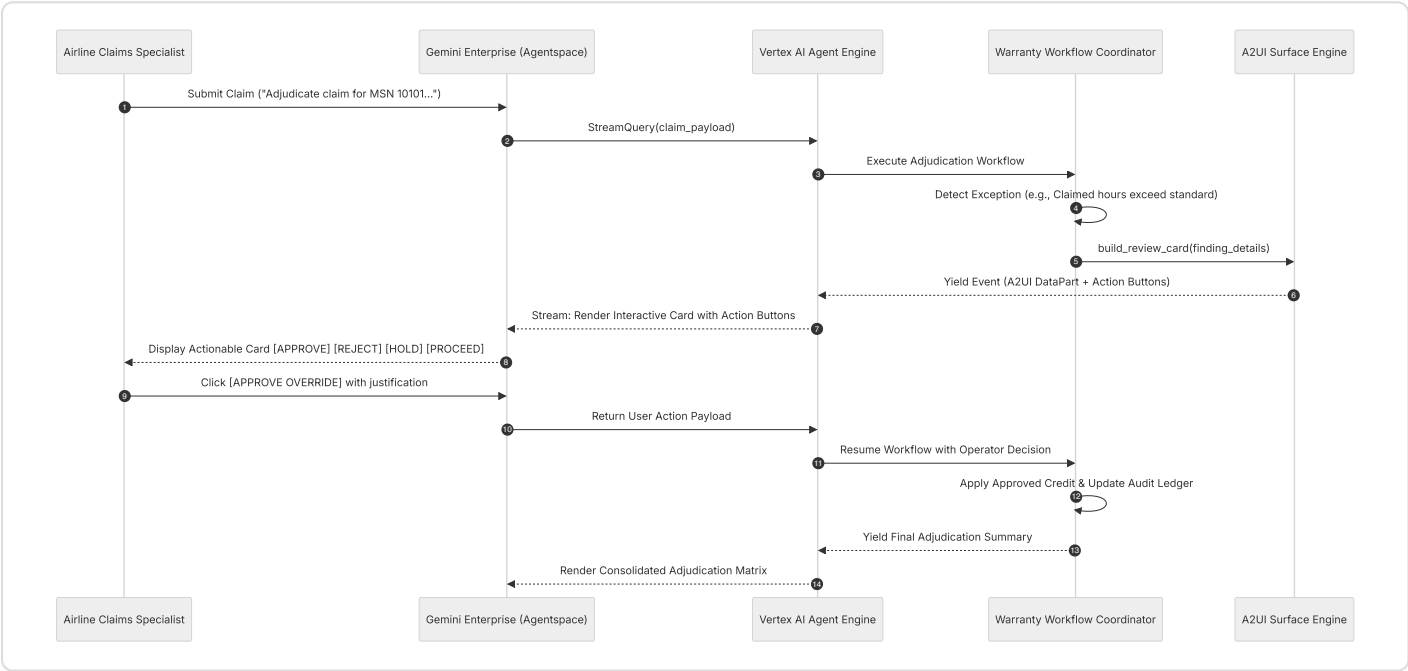
Procedure Hour Suffix Resolver

Service Bulletins designate procedure hours across discrete work packages (**Procedure A** , **Procedure B** , **Procedure C**). Claims specify corresponding dash suffixes (**-01** , **-02** , **-03** , e.g., **SB-DEMO-2801-01**):

- 01** → Standard Procedure A Hours
- 02** → Standard Procedure B Hours
- 03** → Standard Procedure C Hours

4. Human-in-the-Loop (HITL) & A2UI Protocol

To ensure human oversight over financial decisions, the agent incorporates an interactive review surface powered by **A2UI Protocol v0.9**:



The 4 Standard A2UI Decision Variants

- APPROVE_OVERRIDE** : Overrides an automated policy exception with recorded specialist justification.
- PROCEED_STANDARD** : Applies the standard computed credit cap without manual adjustments.
- REJECT** : Rejects non-compliant claims, recording zero credit in the claims database.
- HOLD_FOR_INFO** : Pauses claim processing pending submission of required documentation (e.g., missing task card).

5. Deployment Specifications & Live Production Architecture

SIMULATED RUNTIME DEPLOYMENT MATRIX (SAMPLE)		
Configuration Parameter	Production Deployment Value	Operational Role
Google Cloud Project	`[PROJECT_ID]`	Sovereign host project for Vertex AI & GCS
Vertex AI Runtime	Reasoning Engine (europe-west1)	Containerized agent execution environment
Primary LLM Model	`gemini-3.5-flash`	Multimodal parsing & subagent reasoning
Gemini Enterprise App	Discovery Engine Agentspace	Interactive operator web workspace
Database Ledger	Cloud SQL / BigQuery	Claims audit & historical transaction log
Document Storage	`gs://[CLAIMS_BUCKET]`	Content hashing & attachment archive

Verification & Automated Regression Suite

- Regression Evaluation Set:** Validated via `uv run adk eval` against golden claim test datasets.
- Decision Advisory Framing:** All automated outcomes are presented as **"Suggested for Approval"** with detailed mathematical breakdowns, ensuring specialists retain final authority.